



**NEPAL EARTHQUAKE 2015:
HOUSING RECONSTRUCTION TWO YEARS ON**

**APRIL
2017**



Housing Recovery and Reconstruction Platform

www.hrrpnepal.org

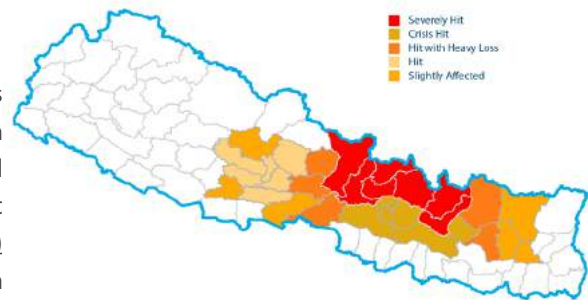
VERSION 1

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INTRODUCTION

The 25th April 2015 Gorkha Earthquake, and the hundreds of aftershocks that followed, including a M6.8 on 12th May, caused loss of life and widespread damage to houses and infrastructure across 31 districts of Nepal. The Government of Nepal (GoN) Post Disaster Needs Assessment (PDNA) categorised these districts as severely hit, crisis hit, hit with heavy losses, hit, and slightly affected.



In the immediate aftermath of the earthquake, emergency shelter assistance was provided to more than 1.2 million houses by GoN, Partner Organisations (POs), private sector, and individuals. This consisted of providing tarpaulins, blankets, and some bedding materials to ensure that affected families could set up emergency shelter. The response then progressed to addressing temporary shelter needs, and the GoN provided, with support from Partner Organisations (POs), a cash grant of 15,000 NPRs to support households with the purchase of materials, or hiring of labour, required to construct a temporary shelter. 699,157 households received the cash grant, or were provided with shelter materials.

The GoN launched an owner-driven housing reconstruction programme early on, and planning for this moved in parallel to the shelter response. The programme was developed based on the key principles set out in the PDNA, in June 2015, which state that reconstruction should:

- Empower communities to take control of their recovery, facilitated through the ODR approach.
- Apply “integrated safer settlement” principles where appropriate, involving the principles of holistic habitat development with an emphasis on basic services and community infrastructure.
- Become a vehicle for building long-term community resilience.
- Strengthen the local economy through processes supportive of the poor, marginalized, and informal sector. It should provide an opportunity for the poor to upgrade their overall living and economic conditions.
- Ensure sustainable and environmentally conscious processes that keep in mind issues such as climate change, natural resource management and scientific risk assessments.
- Be equitable and inclusive, with equal rights to land and property accorded to women.
- Target strategies to address the specific needs of the diverse communities and settlements affected by the earthquakes.

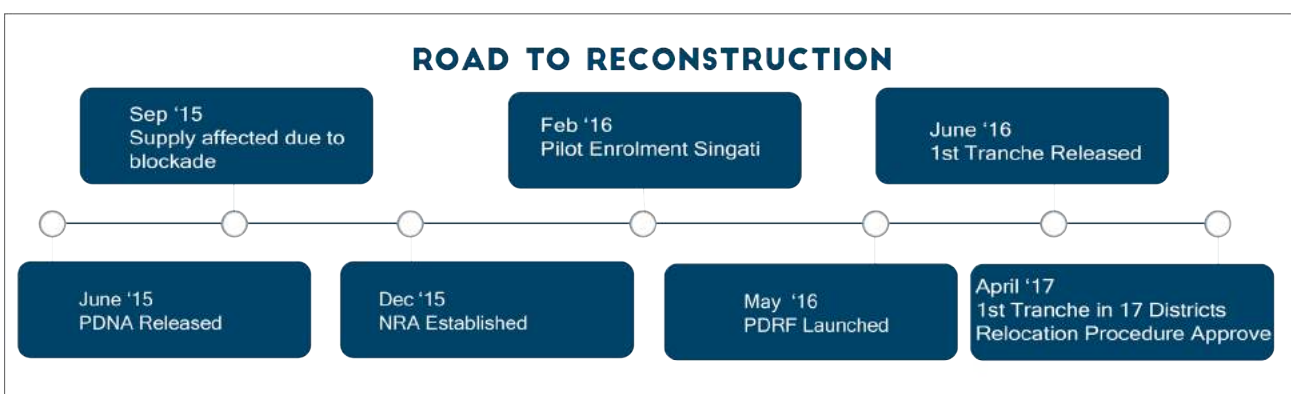
The housing reconstruction programme combines financial assistance from the GoN – in the form of a 300,000 NPRs grant, provided in three tranches linked to compliant construction. The first tranche of 50,000 NPRs is provided once the household has enrolled in the programme, the second tranche of 150,000 NPRs is provided after approval of construction to plinth level, and the third tranche of 100,000 NPRs is provided after approval of construction to ring beam level. A retrofitting grant of 100,000 NPRs is also available to households whose homes have been partially damaged. To complement the grant, socio-technical assistance by the GoN and Partner Organisations (POs) is required to support households throughout the reconstruction. The PDNA calls for “a cascading socio-technical facilitation mechanism for recovery support at the national, district and local level”, which when combined with the financial grant from the GoN will “empower households to lead their own recovery efforts” in compliance with “safe construction standards”.

The National Reconstruction Authority (NRA), formed after the earthquake, has overall government responsibility for the earthquake reconstruction; coordinating, guiding, and supporting the activities of concerned line ministries and local authorities as well as Partner Organisations (POs). The primary responsibility for the disbursement of the housing grant is with the Ministry of Federal Affairs and Local Development (MoFALD). The Ministry of Urban Development (MoUD) and the Department of Urban Development and Building Construction (DUDBC) have primary responsibility for technical standards and compliance. Both ministries implement their reconstruction and recovery activities through Central and District Level Programme Implementation Units (CL-PIU and DL-PIU).

Under the MoUD DL-PIUs more than 2,000 engineers, sub-engineers, and assistant sub-engineers have been deployed to the Village Development Committees (VDCs) to provide technical assistance and conduct inspections, as per the [Inspection Guidelines](#) developed by the NRA. To date, 13,331 houses have been approved for the second tranche of the grant, and 557 have been approved for the third tranche of the grant.

79 Partner Organisations (POs) are providing a range of support to the housing reconstruction across 468 VDCs (out of a total of 618 VDCs, in the 14 districts. In terms of the housing reconstruction, the role of POs, as per the reconstruction policies and guidelines, is to support the delivery of socio-technical assistance at household level. As per the Inspection Guidelines, technical assistance teams at VDC level should, where possible, be formed by POs, MoUD DL-PIU engineers, and MoFALD VDC staff. The technical assistance role optimises the field presence of POs and their skills in outreach and support to vulnerable groups.

SUCCESSSES, CHALLENGES & EMERGING ISSUES



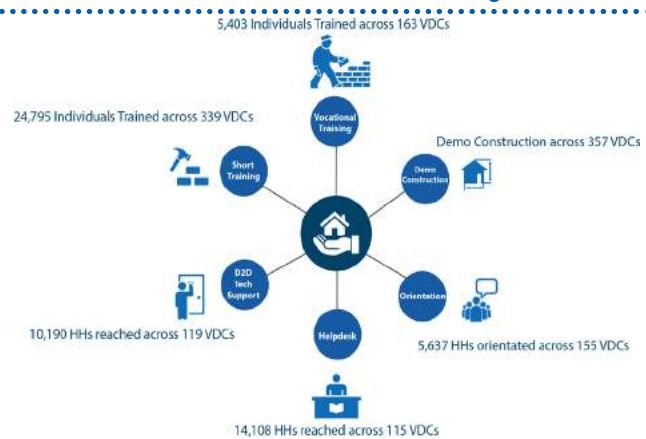
The overriding success is the equitable support programme for all destroyed houses, in all areas, and the establishment of the procedures, guidelines, and systems to operationalise this. Almost 545,000 households, of the 626,695 identified as eligible in the 14 districts, have received the first tranche of 50,000 NPRs which has reassured people that they will access the grant. [The Common Feedback Project \(CFP\) report on reconstruction from February 2017](#) found that “overall, the earthquake affected communities are more positive about reconstruction efforts” primarily as a result of seeing the first tranche being disbursed.

Temporary shelter support: providing a cash grant, and or material support, to households for temporary shelter meant that families could arrange their shelters to fit with their needs, including to have covered space for livestock, storage space, and to build to the required size to accommodate their family and their salvaged belongings.

Detailed damage assessment: a detailed damage assessment has been carried out across the 31 affected districts, establishing a single database of registered households linked to the system for dispersing the grant, and ensuring security and transparency. In addition, a grievance system has been established and of the more than 200,000 grievances registered to date, almost 95,000 have been dealt with. Common grievances include not being listed as eligible for the grant, name appearing incorrectly on the list of eligible houses, not having land ownership documents, and having been missed out of the damage assessment.

Coverage of socio-technical assistance: as reported

in the March 25th 2017, [HRRP 4W](#), POs achievements across the six activities in the technical assistance package include training of over 30,000 masons and construction of more than 350 demonstration houses. The data shows that out of 618 VDCs in the 14 districts, only 25 (4%) are receiving the 6 TA activities. There has been a decrease in the number of VDCs that have no technical assistance, down from 296 in November 2016 to 150 in March 2017, and this represents a positive overall trend in scaling up of TA activities.



Quality of socio-technical assistance: it is not simply a question of rolling out the TA package everywhere, but also about considering how TA teams can be better supported to be effective outreach workers, and what communications methods work best. [BBC Media Action research into communication methods](#) found that printed materials have little or no impact, but family, friends, local chiefs, and / or village elders are the key, and most trusted, sources of information. There is also a lack of common understanding on policies, procedures, technical standards, etc. at village, district, and national level and this is leading to misinformation and confusion.

Funding for socio-technical assistance: 14 POs have committed resources to providing the GoN housing reconstruction grant to a total of 17,457 HHs, or 2.8% of the total number of eligible households in the 14 districts. These funds could have been used, and still could be used, to provide the full package of TA to over 350 VDCs, or nearly 60% of the affected population in the fourteen districts. The funds could also be used to provide top up grants to over 120,000 households. Some Partner Organisations (POs) have already worked with their donor(s) and the GoN to move resources from providing the housing reconstruction grant to expanding socio-technical assistance in their working areas.

Technical standards: the NRA technical working committee recently finalised the 'Corrections and Exceptions' manual which has been designed to support the MoUD DL-PIU engineers with the process of identifying correction measures required for specific non-compliances. This will also be converted into the inspection process, and will support the move away from a 'yes / no' compliance system to a scoring system which reflects and accommodates allowable tolerances. At district level [orientations on the inspection guidelines](#) and [technical workshops](#) have been very useful.

Emerging technical trends: hollow concrete blocks, steel frames, timber structures, hybrid structures, and other emerging trends need relevant technical solutions to support compliance and safe use in construction. New technologies and specifications will need associated training and demonstration. [Volume 2 of the DUDBC design catalogue](#) is a step towards this, but won't impact on compliance until the inspection guidelines are updated. The corrections and exceptions manual will also have to be updated to include these materials and typologies. Core standards were covered early in the recovery, but there is a need to cover additional issues, for traditional and new construction materials, and to address typology issues including numbers of storeys, room sizes, etc. Standards are also required for retrofitting.

One Housing Rehabilitation and Reconstruction Programme, not a collection of approved PO Projects!

The HRRP advocates for contributions and activities of POs to have a common identity, with shared goals and responsibilities, agreed standards, encompassing a range of actors and activities harnessed under the Government-led housing recovery programme. In the absence of this common identity and approach, fragmentation, uneven levels of assistance, gaps where there is no assistance, confusing and contradictory advice, inefficient individual efforts to develop tools, and limited information on what is being provided, where and to whom. There is a great opportunity through the GoN housing reconstruction programme for POs to add value in key roles and work together with central and local officials. We need to work together!

RECONSTRUCTION CONTEXT

The reconstruction context covers a large geographical area and many socio-cultural contexts, with a range of different housing vernacular architecture styles, and a number of emerging housing typologies. This section aims to present an overview of the main contexts.

MID HILLS



DISTRICT:

Gorkha, Dhading, Nuwakot, Lalitpur, Bhaktapur, Kathmandu, Sindhupalchowk, Kavre, Dolakha, and Ramechhap.

NO. OF ELIGIBLE HOUSEHOLD:

511,650 HHs

HOUSING TYPOLOGIES

2.5 storey stone or brick and mud mortar houses, multi-storey RCC frame, hollow concrete blocks, hybrid construction (ground floor masonry, first floor CGI and timber), and steel frame structures.

ISSUES AND TRENDS

Two and half storey homes:

Two and half storey stone masonry homes: communities repeatedly raise concerns about not being able to reconstruct their traditional stone masonry 2.5 storey homes because they do not meet the inspection standards. Traditional homes are large, with space for livestock on the ground floor, and storage in the attic. Technical solutions required to strengthen these traditional buildings so that people can remain in them if appropriate.

Non-compliance: Scope to scale up of technical assistance, technical solutions for 2.5 storey stone masonry. Input needed to ensure compliance criteria is updated to reflect households needs and address emerging issues for inspection – started with corrections / exceptions manual for households to rectify basic non-compliance.

Road Widening: uncertainty about plans for road widening affect many of the communities in these areas and as a result they are reluctant to start construction of their homes as they do not know what impact road widening will have on their plot, both in terms of size and what, if any, financial support they will receive from the GoN if their land is part of road widening.



Trained masons working on the construction of a house in ward three of Bijulikot VDC, Ramechhap. Photo credit: Medair

¹ <http://202.45.144.197/nfdnfs/clpiu/index.htm>



Construction using hollow concrete block in Nalang VDC, Dhading.

SOUTHERN DISTRICTS



DISTRICTS:

Makwanpur, Sindhuli, Okhaldhunga

NO. OF ELIGIBLE HOUSEHOLDS:

84,326 HHs

HOUSING TYPOLOGIES

stone or brick masonry with partial damage and potential for retrofitting, timber structures, bamboo, concrete block, and hybrid

ISSUES AND TRENDS

Retrofitting: how can the 100,000 NPRs grant be operationalised and what will the inspection process for this look like? How will technical assistance / engineering support be managed? Retrofitting requires very strong technical support which may be difficult at scale.

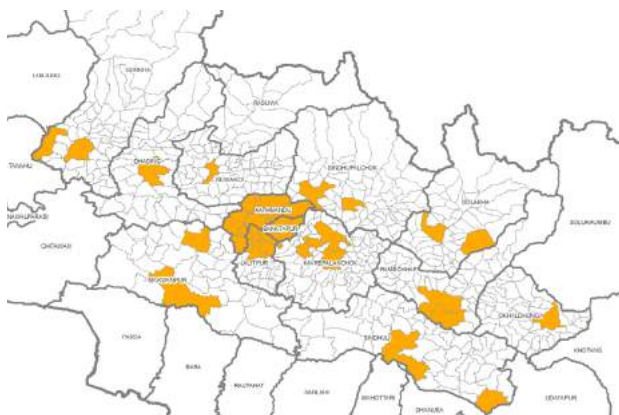
Lack of resources: the coverage of technical assistance is very low in many of these areas. Significant gaps and lack of partner presence.

Timber houses: the extensive use of timber in construction in this region requires clarifications on standards and guidance to ensure appropriate selection, treatment and use, safe structural design and construction.



The NRA Chief in Okhaldhunga monitoring work in Harkapur VDC.

URBAN AREAS



DISTRICTS:

Kathmandu, Bhaktapur, Lalitpur, 42 municipalities in other earthquake affected districts

NO. OF ELIGIBLE HOUSEHOLDS:

172,015 HHs

HOUSING TYPOLOGIES:

traditional Newari architecture with brick masonry and intricate wood carving on openings built around courtyards, often mixed use buildings with commercial units on ground floor and residential above, large RCC multi-storey buildings. Smaller RCC buildings with brick infill. Concrete blocks.

ISSUES AND TRENDS:

Socio-technical assistance needs to be delivered using different a methodology from rural areas and blanket coverage is not feasible. The TA should be delivered through, and support existing municipal mechanisms. Fostering collaboration between the municipal engineers and the MoUD DL-PIU engineers is also important. Private sector, consulting engineers, contractors, and local universities play an important role also, and there are advantages to having access to experienced and skilled engineers and architects but households may benefit from support to identify who these people are, such as a roster of expertise. Continued professional development for engineers and architects is also important.



courtyard in Patan

Urban reconstruction and rehabilitation requires a range of guidance on conservation and appropriate upgrading works, services and reuse to ensure cultural, social and economic viability along with physical safety. Advice, policy measures and support are needed for individual buildings, for blocks of buildings, neighbourhoods and villages. Technical issues and trends include the high number of small plots due to the practice of vertical subdivision when home ownership is transferred through family members. This leads to very narrow, slender houses which cannot be constructed as per NBC. Traditional buildings are often replaced with RCC frame structures which is having an impact on heritage preservation and may have an impact on tourism. Retrofitting may be an option for many houses in urban areas, but there are mixed levels of expertise and understanding on this amongst engineers and contractors and as a result the quality of retrofitting approaches varies.

Municipal by-laws: the application of the compliance criteria as set out in the inspection guidelines can be complicated because of contradictions between compliance standards and the standards set out in municipal by-laws.

Renters are numerous in municipalities and urban areas. The approach to reconstruction needs to be inclusive of renters. Many displaced families in urban areas were renters before the earthquake.

Financial measures: construction in urban areas is generally much more expensive than in rural areas, this means that households in urban areas will need access to appropriate financial measures and information to manage their construction.

NORTHERN AREAS



DISTRICTS:

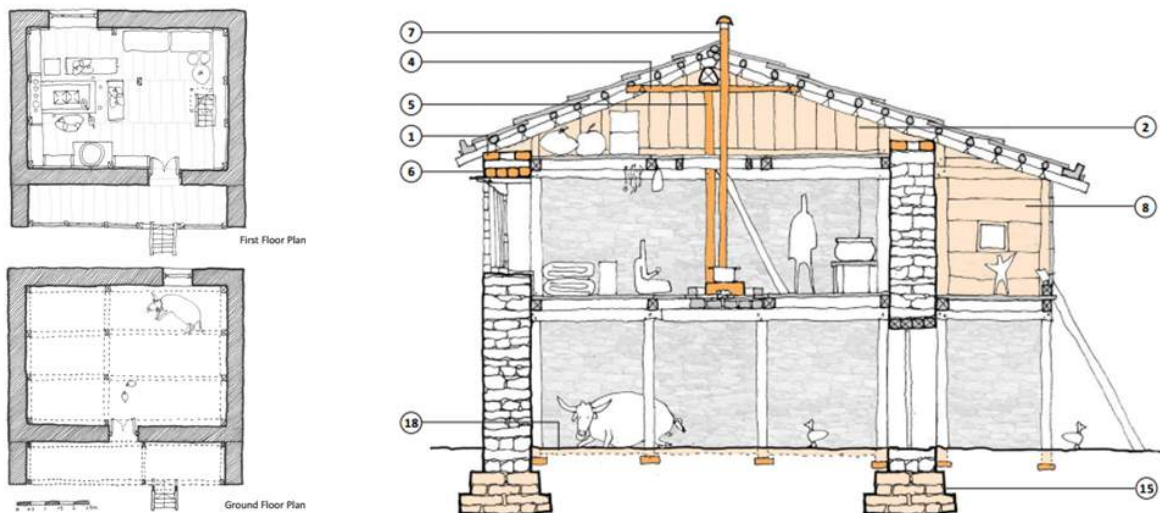
Gorkha, Dhading, Rasuwa, Sindhupalchowk, Dolakha, and Ramechhap.

NO. OF ELIGIBLE HOUSEHOLD:

26,959m above 2,000m, 3,760 above 2,500m

HOUSING TYPOLOGIES:

2.5 storey dry stone masonry; timber and masonry mix



Floor plan and section, typical Khim house, Gorkha³



Gatlang VDC, Rasuwa, June 2015

Photo Credit: GOAL

³ 4th April Northern Areas Technical Session, HRRP presentation, <http://www.hrrpnepal.org/meeting-detail/view/northern-areas-technical-session>

ISSUES AND TRENDS

Defining remote northern areas:

currently looking at altitude, road access, and access to markets as the primary factors. The definition will have implications on the number of households that may be eligible to receive a remote top-up grant.

Non-compliant architecture:

Need for identification of common building designs, photograph and draw them, and then have MoUD CL-PIU structural engineers work to determine the relevant inspection criteria and corrections for common non-compliances. Some of these building designs may be permitted in an expanded menu for compliance, after relevant improvements have been incorporated.

Technical assistance approach:

What should model houses demonstrate? How should orientations be delivered, who should deliver them, what should the content be? What training do technical assistance staff need, what languages do they need to be able to speak, how can they be monitored and supported? Can helpdesks provide more than housing advice and where will they be located? With many houses already built the approach to housing reconstruction will be very different.

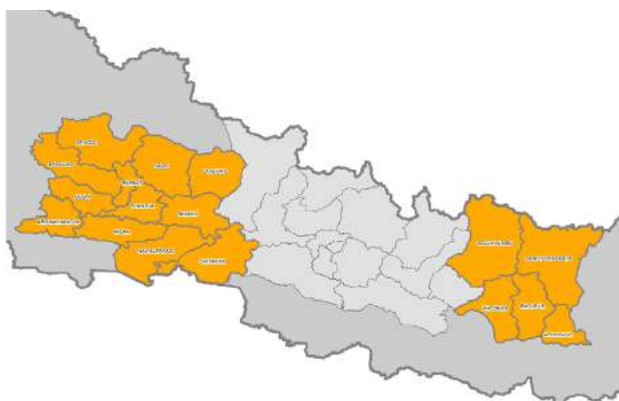
Access:

Communities in northern areas reportedly prioritise improved access over the housing reconstruction when asked about their reconstruction needs, but there is also a fear around what impact this will have as many jobs are tied to portering, mules, trekking, etc. Remote and inaccessible areas also pay more for materials due to the high cost of transportation. Expanding compliance menu to include local technologies will help to address this. Labour supply can also be an issue if people travel for seasonal work or if outside labour is brought in who are not skilled enough and who don't know the traditional building practices.

Tourism:

With many trekking routes in the northern areas there is potential for tourism to have a big impact on the reconstruction in these areas. Linking reconstruction and recovery to tourism is also a priority of the Post Disaster Recovery Framework (PDRF)

17 DISTRICTS



DISTRICTS:

Myagdi, Kaski, Lamjung, Baglung, Parbat, Syanja, Tanahun, Gulmi, Arghakhanchi, Palpa, Nawalparasi, and Chitwan, Solukhumbhu, Sankhuwasabha, Khotang, Bhojpur, and Dhankuta

NO. OF ELIGIBLE HOUSEHOLD:

91,677 HHs⁴

HOUSING TYPOLOGIES

Same as 'Mid-Hills' in Kaski, Lamjung, Baglung, Parbat, Synagja, Gulmi, Argakhanchi and Palpa. Same as 'Northern Areas' in Myagdi, Solukhumbhu, Sankhuwasabha, Khotang, Bhojpur and Dhankuta. Same as 'Southern Areas' in Nawalparasi and Chitwan.

³ Gorkha North Field Report, CRS Shelter and Settlements Team, July 2016 http://www.hrrpnepal.org/download/olgtzAFF4sNPqH71t58k_2017_04_04.pdf/meeting

ISSUES AND TRENDS

Corrections:

First tranche disbursement launched 3rd April, starting in Kaski district. It is expected that many households will have already built / repaired so there will be need for corrections which requires strong technical support.

Retrofitting:

There are expected to be many houses that may be retrofitted, rather than fully reconstructed, as the level of damage in the 17 districts is generally lower than in the 14. How can the 100,000 NPRs grant be operationalised and what will the inspection criteria and standards be. How will technical assistance / engineering support be managed? Retrofitting requires very strong technical support which may be difficult at scale.

Programme management structure:

It is still not clear if there will be an NRA office in each district, or if there will be District Level Programme Implementation Units (DL-PIUs) under MoUD and MoFALD.

Technical assistance:

What should the approach be, and what lessons have been learnt from providing technical assistance in the 14 districts that can be taken into the 17. The structure to manage the role out of TA will be different due to the number of POs present.

Geographical spread:

The districts themselves are spread across an enormous area, and then within the districts the eligible households are spread across a large distance, this will make logistics more challenging for technical assistance and inspection.



Typical house in Arghakhanchi district



Typical Tharu house in Nawalparasi district

⁴ As of 21st April 2017

DISPLACED HHS



DISTRICTS

Sindhupalchowk, Rasuwa, Gorkha, Bhaktapur, Kathmandu, Lalitpur, Nuwakot, Dolakha, Dhading, Makwanpur, and Ramechhap⁵.

NO. OF ELIGIBLE HOUSEHOLDS

[Round nine of the Displacement Tracking Matrix \(DTM\)](#), identified 3,753 displaced households (15,595 people) living across 65 sites.

ISSUES AND TRENDS

Procedures for the Relocation and Rehabilitation of Hazard-prone Settlements, 2073 (2017):

On 7th April, the NRA Executive Committee approved the procedure under which, households whose land has been identified as at risk can choose to relocate either within their own district or in any other earthquake-affected district close by. A GoN grant will be provided for the purchase of land. Tsujan how the land ownership certificate must be issued jointly where the purchase is made by a couple, and the ownership of the land cannot be transferred for 10 years. The procedure encourages displaced families to create users groups involving at least 10 families from the settlement so that they can jointly select a safe location for an integrated settlement. Displaced families may relocate individually if that is their preference and are free to choose their preferred options for new locations.

Geo-Hazard risk assessment:

Conducted by the National Reconstruction Authority (NRA) in more than 500 locations across 15 districts to assess and categorise sites as either category I: safe communities/villages where reconstruction can be started any time; category II: communities/villages under the risk of manageable geo-hazards where reconstruction can be started only after applying suitable mitigation measures; and category III: unsafe communities/villages due to the existing state of geo-hazards which are extremely difficult to control technically as well as financially and where reconstruction is not recommended. The assessment also identified potential relocation sites where required. The relocation procedure defines those families in category III to be eligible for the relocation grant, whether or not they have been listed as eligible for the housing reconstruction grant under the damage assessment.

⁵ DTM Round 9, <https://drive.google.com/file/d/0B3CEVcVlpExQsJvGYU5Vd2ljR0k/view>



Photo credit, Steve Revill

Local displacement:

Round 9 of the DTM found that 49% of displacement sites are within 30 minutes from the displaced families place of origin, 26% are between 30 minutes and 3 hours from the place of origin, and 25% are more than 3 hours from the place of origin. The prevalence of local displacement means that many of the families living in displacement sites have not lost access to services or jobs.

Eviction:

On 14th March 2017, the largest displacement site in Kathmandu was cleared by Nepal police following an eviction notice which had been issued to residents at the site. 95% of residents at the site were from the local area and many had been renting prior to the earthquake and were unable to find rental accommodation at the same rate after the earthquake. 5% of residents were from outside Kathmandu and had moved to the site as their land had been affected by landslides. Eviction remains a threat for some displacement sites.

Monsoon preparedness and potential impact:

High potential for flooding / rainfall to impact on families still living in displacement sites. The Durable Solutions consortium has prepared a severity index which should support with prioritising mitigation works to increase preparedness and resilience in advance of the monsoon. The Shelter Cluster contingency plan has been updated, but indication seems to be that resources available for monsoon response are limited.

Key Ways Forward

Some of the key ways forward for the next years of the housing reconstruction include:

- Continue to build standards and solutions to address technical issues coming from the districts
- Document building typologies, materials, rate, etc.
- Increase coverage of technical assistance by redistributing funds originally committed for the reconstruction housing grant to technical assistance activities
- Expand coverage of the vulnerable top-up grant by redistributing funds originally committed for the reconstruction housing grant
- Expand technical assistance support to 17 districts and develop relevant standards and technical solutions related to the housing typologies across these areas
- Support VDC Secretaries to activate community reconstruction committees to ensure
- Work with, and support DL-PIU engineers and VDC staff to build their capacity and strengthen their roles so that they can effectively manage their tasks
- Engagement of, and support to, masons as key actors in the reconstruction
- Equip technical assistance staff with the skills to communicate effectively with HHs and communities and to act as outreach workers
- Develop market hubs as information hubs, and strengthen the role of producers / fabricators / vendors of construction materials to be able to provide information to home owners and to improve quality of materials
- Monthly orientations for GoN and PO field staff, improve collaboration, ensure common understanding and approach, as well as information flow from field level to national and vice versa, and provide opportunities for capacity development
- Move towards a more integrated housing response to meet the indicators set out in the Post Disaster Recovery Framework

The Housing Recovery and Reconstruction Platform (HRRP) exists to support the National Reconstruction Authority (NRA), Ministry of Urban Development (MoUD) and Ministry of Federal Affairs and Local Development (MoFALD) Central Level Programme Implementation Units (CL-PIUs), other relevant government authorities, and Partner Organisations (POs) with coordination of the post-earthquake housing reconstruction.

HRRP is supported by DFID, CRS Nepal, NSET, SDC, Caritas Nepal, Build Change, ACTED, Plan International, and Habitat for Humanity, and has been approved to run until the end of February 2019.

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